ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY **9189/4**

PAPER 4 Option Topics

NOVEMBER 2015 SESSION

1 hour 15 minutes

Additional materials:
- Answer paper
- Data booklet
- Graph paper
- Mathematical tables and/or electronic calculator

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper.

Answer a total of **four** questions. Do not answer more than **two** questions from any **one** Option.

You are advised not to attempt questions on Options for which you have not been prepared.

Write your answers on the separate answer paper provided.

Begin each answer on a fresh page.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

Where relevant, the symbols for aluminium, chlorine and iodine are printed as: A*l*; C*l*; I; respectively.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 11 printed pages and 1 blank page.

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BIOCHEMISTRY

Not more than two questions to be answered from this option.

1 In a polymerase chain reaction, DNA is copied a million times by placing in the reaction vessel a DNA strand, ATP and a polymerase enzyme at an appropriate temperature.

(a) **(i)** Name the process of copying DNA. 1

(ii) Explain the process of copying a DNA strand during protein synthesis. 2

(iii) State the product of the process in **(ii)** and site of its occurrence. 2

[5]

(b) The sequence of a DNA molecule can be read as a triplet code.

(i) Define the term *triplet code.* 1

(ii) Name the process of reading the triplet code and state the site where the process occurs. 2

[3]

(c) **(i)** State the role of ATP in living organisms. 1

(ii) Explain why ATP is easily hydrolysed? 1

[2]
[Total:10]

2 **(a)** **(i)** State any **two** functions of triacylglycerol in living organisms.

(ii) Write a general chemical equation for the formation of triacylglycerol from glycerol and a carboxylic acid of the general formula RCOOH.

[4]

(b) **Fig.2.1** shows the Na^+ - K^+ pump of animal cells.

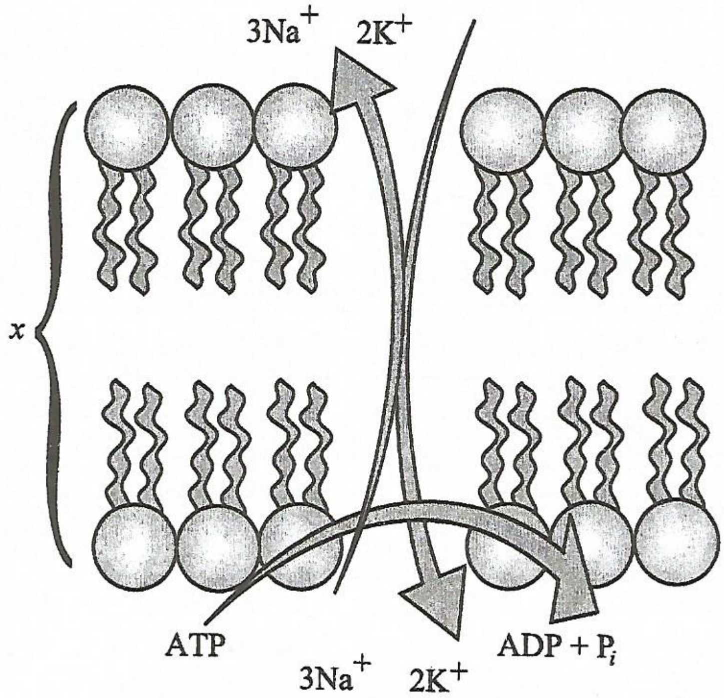

Fig.2.1

(i) Name the part labelled *x*.

(ii) State the functions of part *x*.

(iii) Explain the role of the Na^+ - K^+ pump in living cells.

[6]
[Total:10]

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3 (a) (i) Define the term *cofactor*.

(ii) Give **one** example of a *cofactor*.

[2]

(b) **Table 3.1** shows how the initial velocity of an enzyme controlled reaction is affected by the presence or absence of molecules **A** and **B**.

[S]/M	velocity/ $Ms^{-1} \times 10^{-7}$	velocity with A $Ms^{-1} \times 10^{-7}$	velocity with B $Ms^{-1} \times 10^{-7}$
0.33	1.65	1.03	0.79
0.40	1.86	1.21	0.89
0.50	2.13	1.43	1.02
0.67	2.13	1.74	1.19
1.00	2.49	2.22	1.43
2.00	2.00	8.08	1.79
2.50	3.72	3.72	1.80

(i) Use the given data to plot graphs on the same axes of velocity of reaction versus concentration.

(ii) determine K_m and V_{max} values.

(iii) Explain the effect of the molecules **A** and **B** on the reaction. [8]

[Total:10]

ENVIRONMENTAL CHEMISTRY

*Not more than **two** questions to be answered from this option.*

4 Hwange thermal power station contributes a great deal of sulphur dioxide into the atmosphere.

(a) **(i)** Give the source of SO_2 at Hwange.

(ii) Write an equation to show how SO_2 is formed.

(iii) Using equations state and explain **two** methods that could be employed in controlling SO_2 emission.

(iv) Describe the effects of SO_2 emission into the atmosphere.

[7]

(b) **(i)** Define the term *incineration.*

(ii) State **one** advantage and **one** disadvantage of incineration.

[3]
[Total:10]

5 **(a)** **(i)** State the role of ozone in the stratosphere. 1

(ii) Using equations explain how the concentration of ozone are maintained. 4

(iii) Describe the effects of ozone depletion. 3

[8]

(b) The equilibrium of gases in the atmosphere depends on their residence time.

Define the term *residence time* and give an example. [2]
[Total:10]

6 (a) (i) Define the term

1. *cation exchange capacity,*

2. *temporary cation exchange.*

(ii) Explain, using examples, how ion substitution can occur in clay soils.

[4]

(b) Suggest, using redox potentials, the role of oxygen in soil in terms of

1. nitrogen compounds,

2. iron species.

[4]

(c) Explain how bed rocks containing limestone affect soil acidity. [2]

[Total:10]

PHASE EQUILIBRIA

*Not more than **two** questions to be answered from this option.*

7 **(a)** Define the term

(i) *partition,*

(ii) *partition coefficient.*

[2]

(b) Butanoic acid of mass 1.20 g was shaken with a mixture of 100 cm^3 of water and 100 cm^3 of ether. After equilibration, the ether layer had 0.024 moles of butanoic acid whilst the water layer had 0.16 moles.

(i) Calculate the partition coefficient for butanoic acid between ether and water.

(ii) State and explain any **two** factors which affect the magnitude of the partition coefficient in **b(i).**

(iii) State, with a reason, whether a mixture of ethanoic acid dissolved in water obeys the partition law when added to benzene.

[8]
[Total:10]

8 **(a)** Two immiscible liquids CCl_4 and CH_3OH were placed in a closed container.

(i) State the number of phases present in the closed container.

(ii) Identify the composition of each phase in the closed container.

[4]

(b) At 1 atm pressure, HCl and H_2O form an azeotropic mixture containing 20.22% HCl, that boils at 109 °C.

Draw

(i) the vapour pressure composition curve for HCl and H_2O,

(ii) the boiling composition curve for HCl and H_2O.

In each case show the liquid curve, the vapour curve and the azeotropic composition.

[4]

(c) Describe how a constant boiling point $HCl_{(aq)}$ solution can be prepared in a laboratory. [2]

[Total:10]

9 **(a)** Define the term *electrophoresis*. [1]

(b) **Fig. 9.1** shows the apparatus used for analysis of amino acids by electrophoresis.

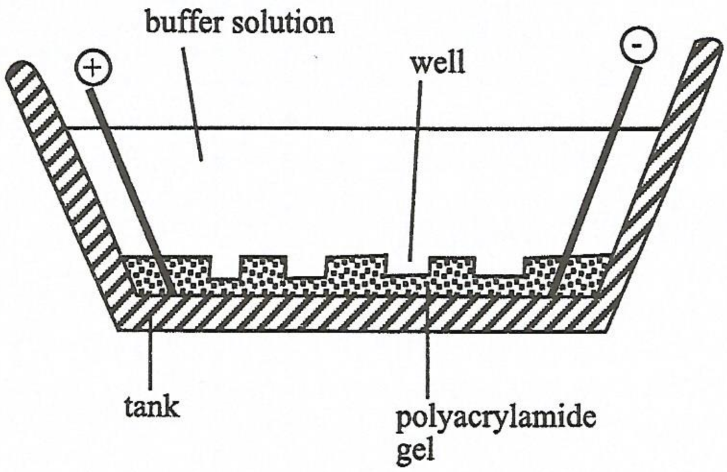

Fig. 9.1

State the function of the

(i) polyacrylamide gel,

(ii) well,

(iii) buffer solution.

[3]

(c) **Fig. 9.2** shows the electrophoresis titration curve for 2- aminoethanoic acid with a strong alkali.

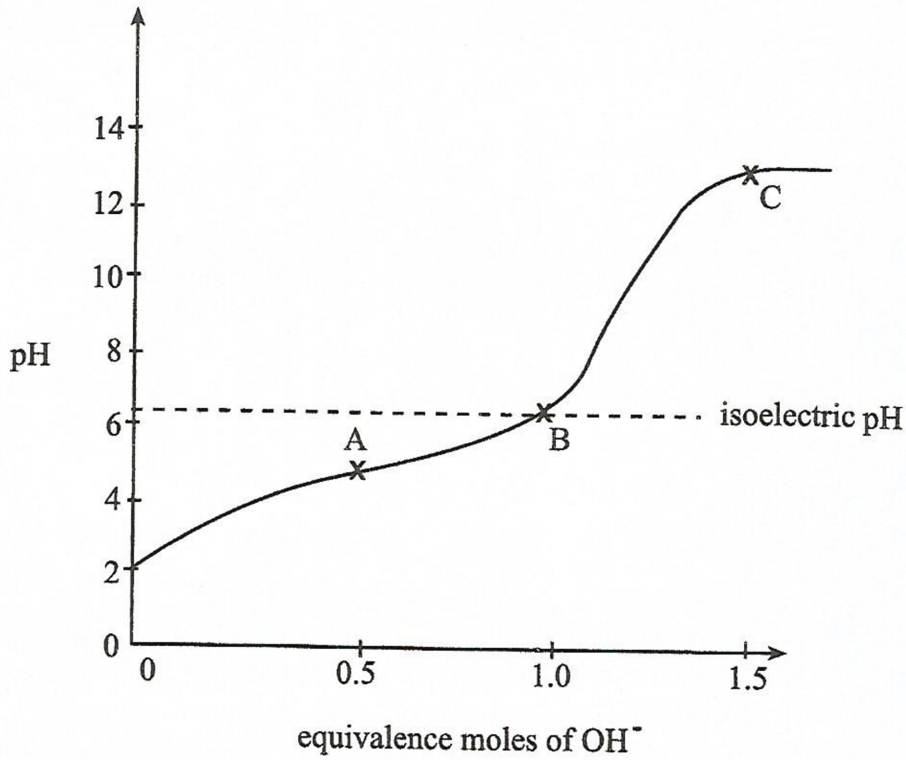

Fig. 9.2

(i) Give the structures of the organic species present in the solution at point A, B and C.

(ii) Describe and explain the movement of each species if electrophoresis was carried out at a pH of 7.

[6]
[Total:10]

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TRANSITION METALS

Not more than two questions to be answered from this option.

10 Oxides of manganese in low oxidation states tend to be basic while in high oxidation states are acidic.

(a) **(i)** State the oxidation state of Mn in

1. MnO,

2. Mn_2O_7.

(iii) Write chemical equations to show

1. the acidic properties of Mn_2O_7,

2. basic properties of MnO.

[4]

(b) The standard electrode potential indicates the relative stabilities of various ions in solution.

(i) Define the term *standard electrode potential.*

(ii) Using E^{θ} values, explain if the reaction between chlorine and $Mn^{2+}_{(aq)}$, to $MnO_{2(s)}$ is feasible. [6]

[Total:10]

11 **(a)** Explain the following observations:

(i) chromium is resistant to corrosion despite its highly negative E^{θ} value

(ii) chromium metal react with aqueous hydrogen ions to give a blue solution. The blue solution quickly turns to a green solution when exposed to air

[4]

(b) Give the colour and chemical equation for the reaction of the complex ion, $\left[Cr\left(H_2O\right)_6\right]^{3+}$ with

(i) excess $NaOH_{(aq)}$,

(ii) $NH_{3(aq)}$.

[3]

(c) (i) Give **one** use of chromium. [1]

(ii) Draw and state the type of isomerism shown by dichlorotetraamine chromium (III) cation, $\left[Cr\left(NH_3\right)_4 Cl_2\right]^+$. [2]

[Total: 10]

12 (a) Predict the shapes of the following complexes.

(i) $\left[V\left(C_2O_4\right)_3\right]^{3-}$

(ii) $VOCl_3$

[2]

(b) (i) Calculate the oxidation state of vanadium in the complexes in **(a)**.

(ii) State, with a reason, whether the complexes in **(a)** are diamagnetic or paramagnetic.

[5]

(c) Write balanced chemical equations for the oxidation for Fe^{2+} by VO_2^+, stating the observable changes. [3]

[Total:10]

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